

Masked Face Completion

Residual Attention Conditional UNET

NCVPRIPG 2023
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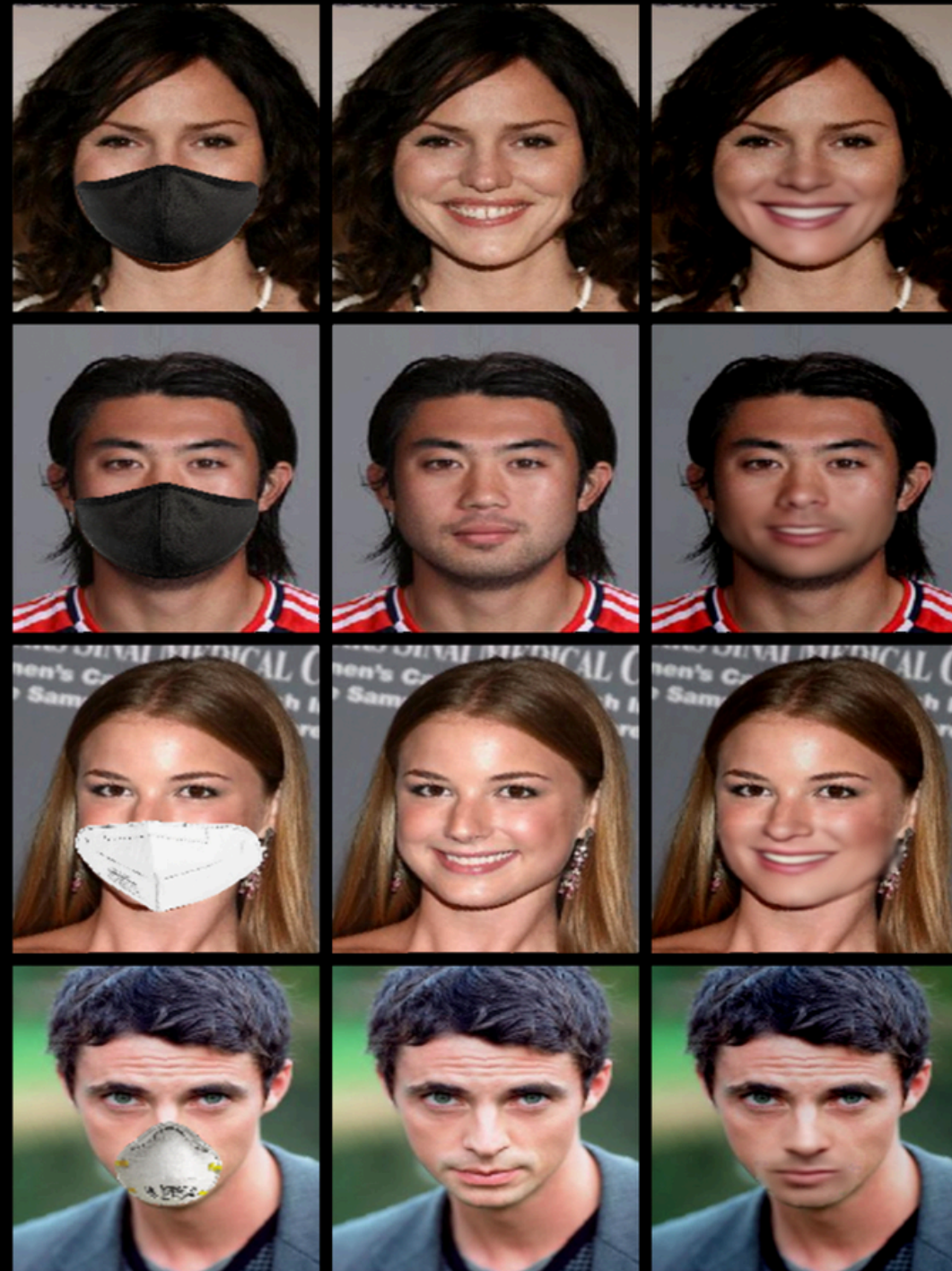
INTRODUCTION

The Covid-19 pandemic has forced people to wear masks and kept us all guessing what the face looks like behind the mask. This makes our security and identification systems vulnerable. Thus, restoring covered face is an interesting problem with significant real-world applications.

INPUT

GT

MODEL



Source: STRV Mask2Face blog

SCOPE OF THE WORK

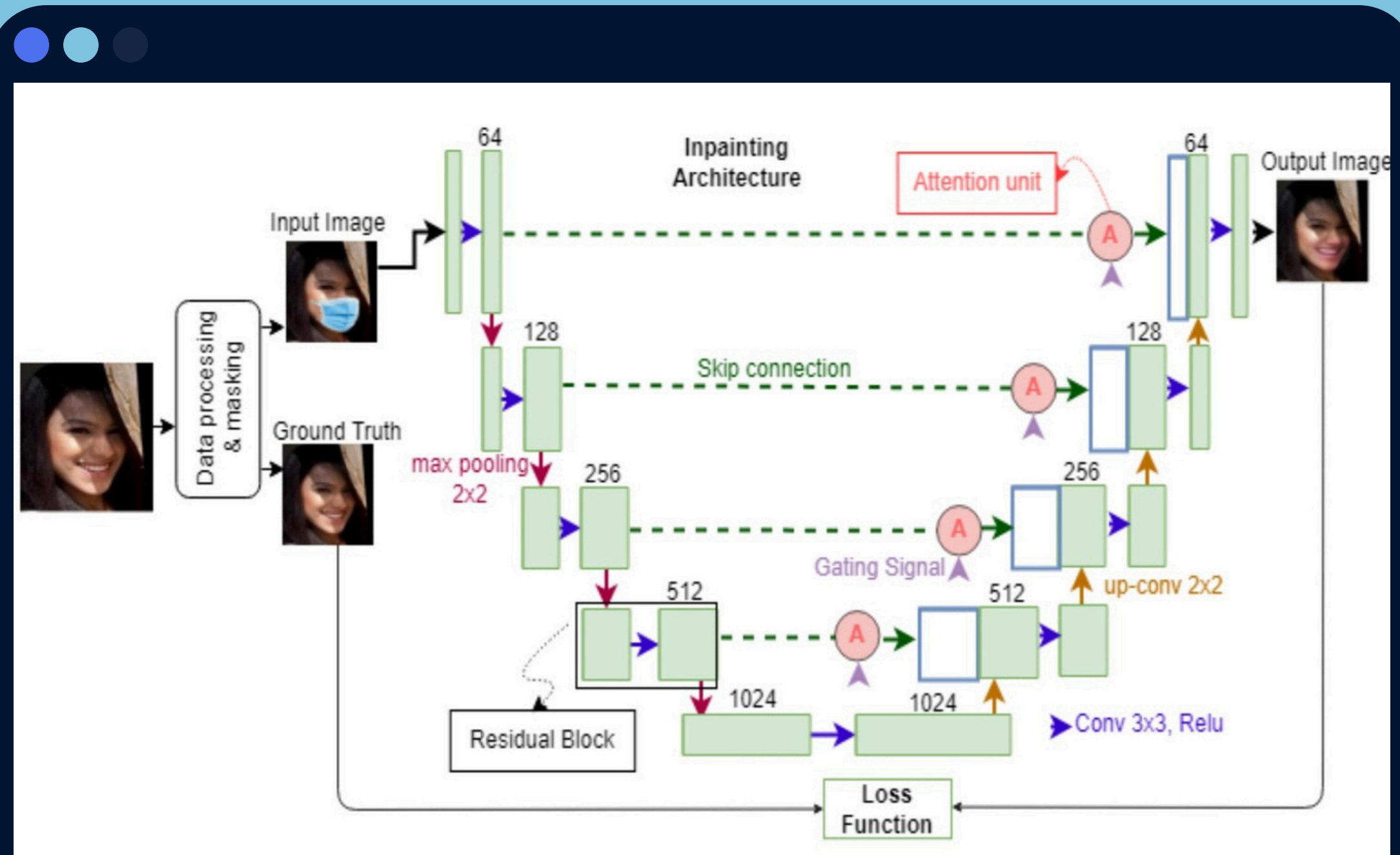
Masked Face Completion for different emotions using Face-Inpainting techniques, using the emotion label as an additional input.

The work proposes the Residual Attention Conditional UNET, Our approach leverages an expression label to generate high-quality images. The model's robustness and accuracy in in-painting faces are also tested.

RESIDUAL ATTENTION UNET

Developed by Md. Hosen et al (2022).

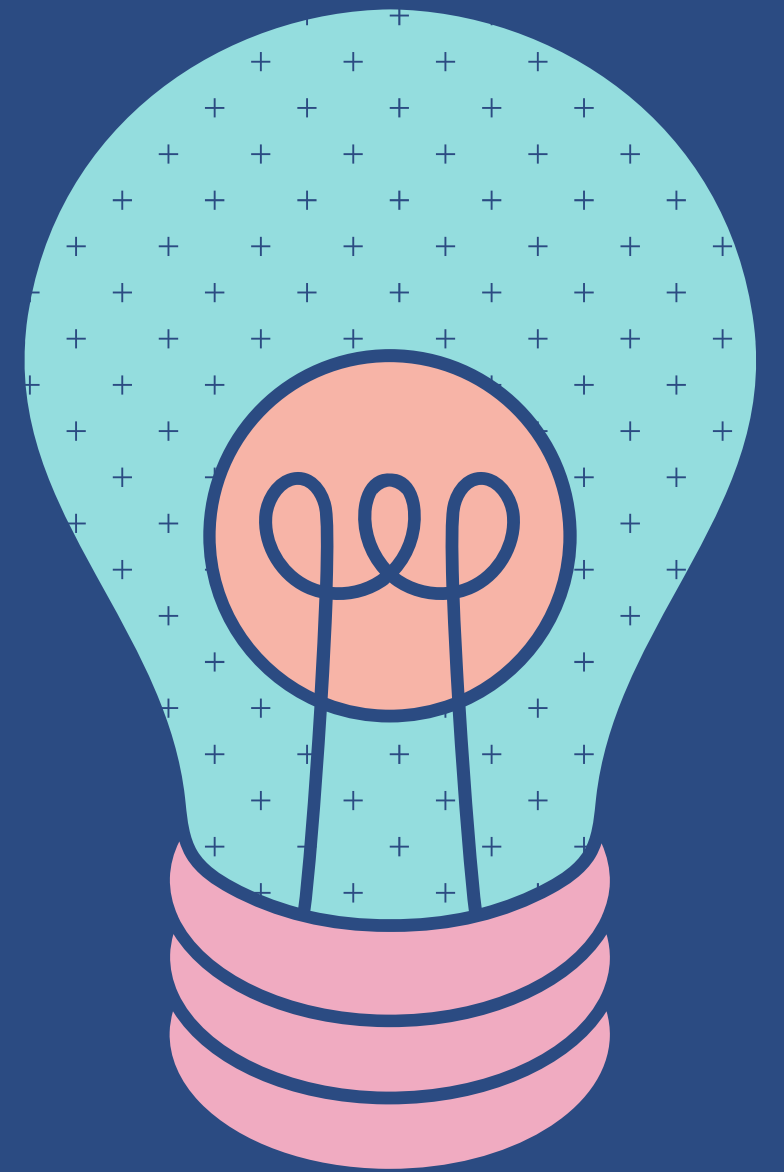
- Blind in-painting technique.
- UNETs were originally used for Biomedical Segmentation.
- Consists of a basic Encoder-Decoder architecture.
- Proposed model used Attention Mechanisms as well as Residual Blocks

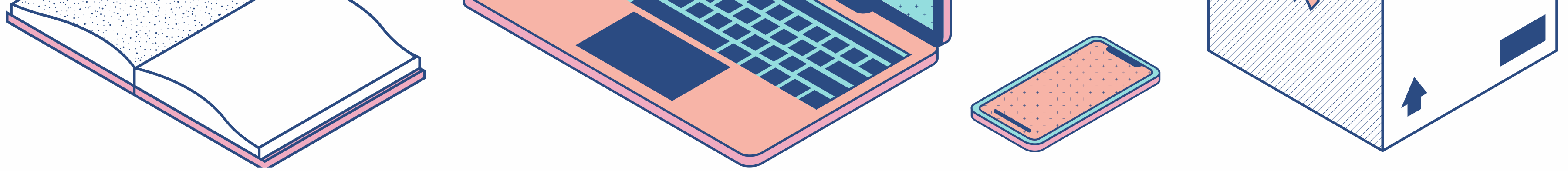


Source: Masked Face Inpainting through Residual Attention U-NET

The model was originally trained on CelebA dataset using SSIM loss with L1 normalization.

**TO GENERATE AN IMAGE WITH THE
ACCURATE REPRODUCTION OF THE
EMOTION, WE NEED TO MAKE USE
OF THE SEMANTIC INFORMATION
WE HAVE.**





Training Setup

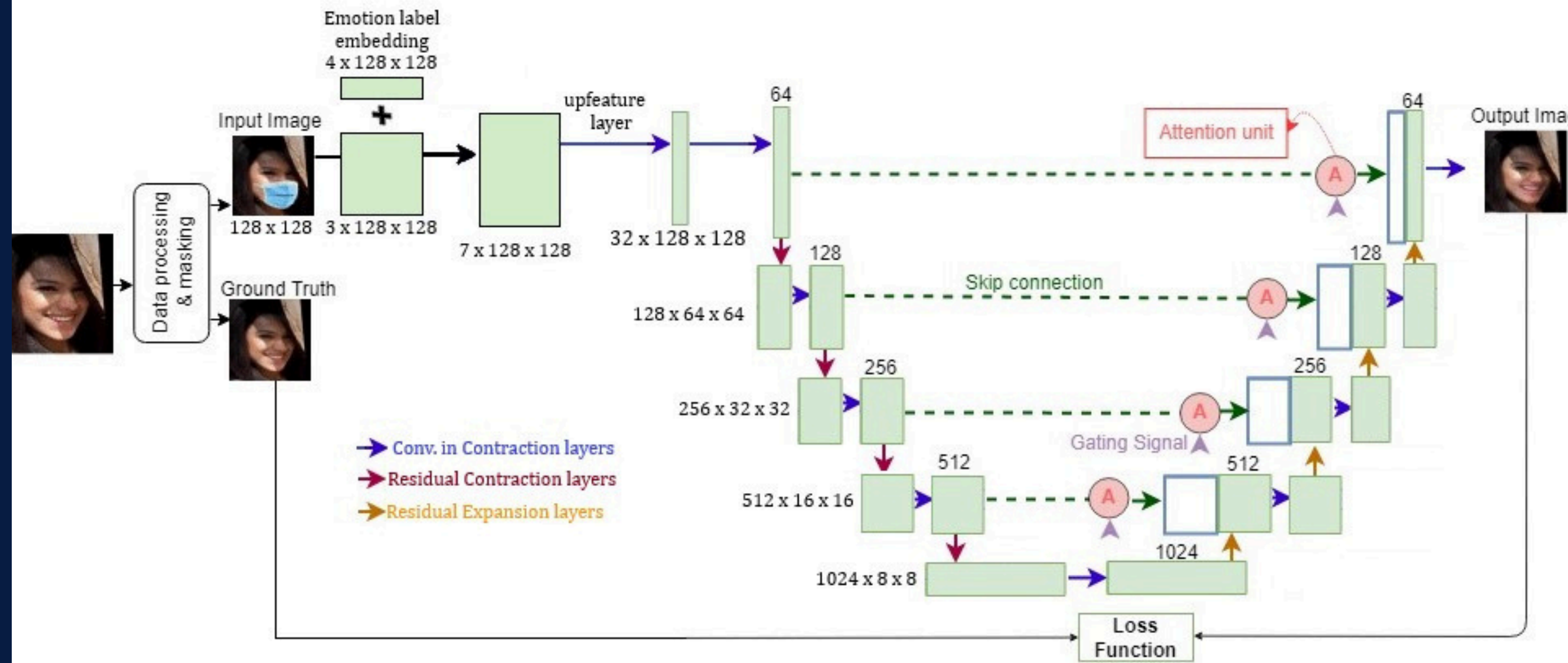
- **Dataset:-** RAFDB (Real-world Affective Faces Database) with **12271** samples for Training and **3067** for validation.
- The dataset was prepared for the training by running a script to mask the faces digitally called **MaskTheFace** by **Aqeel Anwar**.
- Another dataset called **AffectNet** also was later utilized to augment and balance the dataset.
- Loss function primarily used: **MS-SSIM** (Multi Scale Structural Similarity index)
Batch Size: 8
Optimizer: **Adam** with initial lr = **0.0001**
- The work was entirely done on **Kaggle** and their Tesla P100 GPUs.



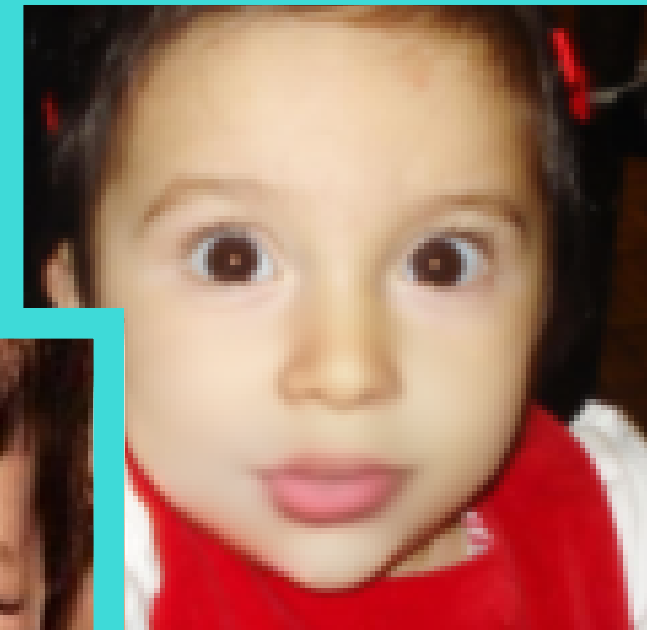
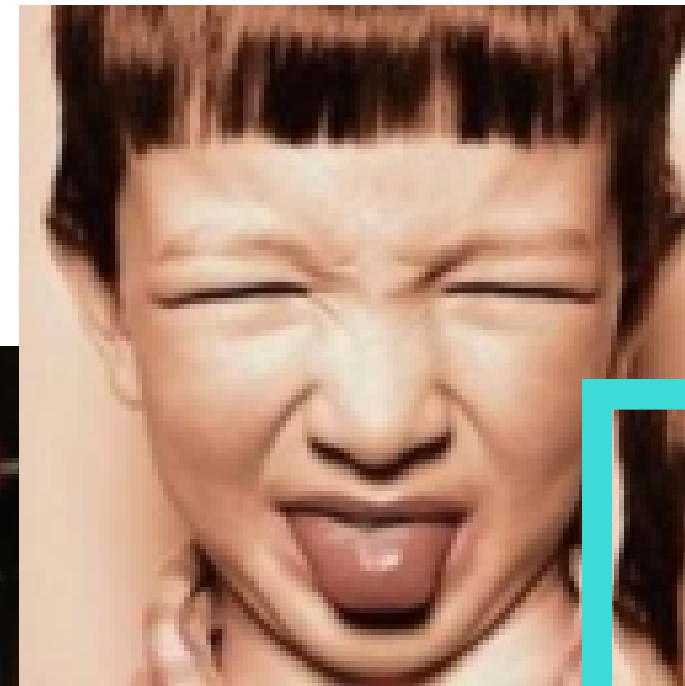
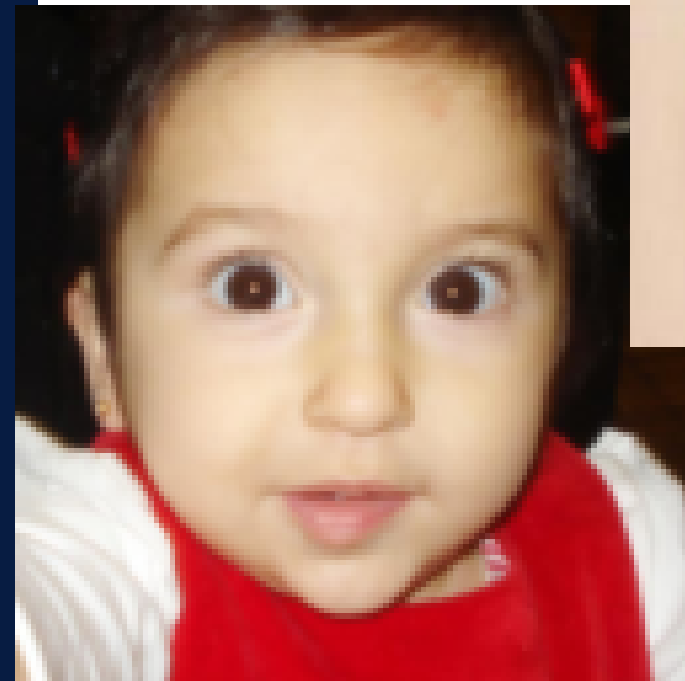
RAC-UNET

Residual Attention Conditional UNET (RAC-UNET) is an improved model that uses Embedding layers to feed the emotion input.

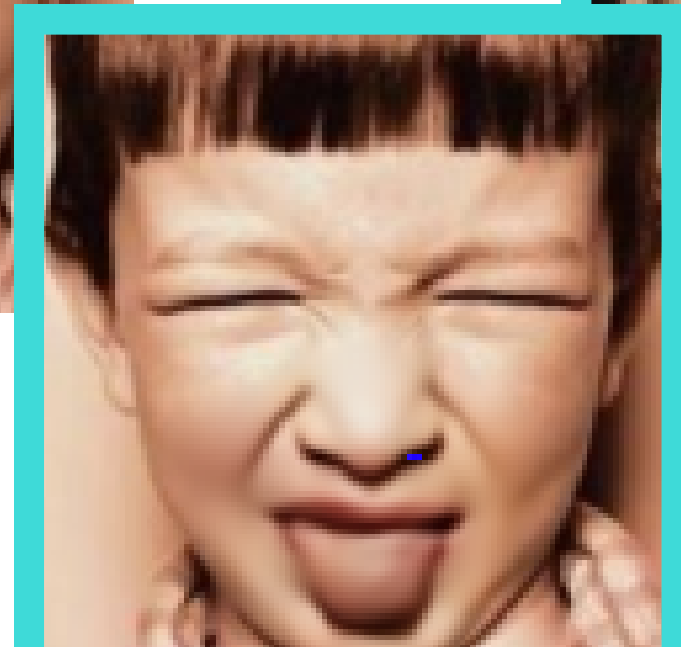
It also has custom Residual blocks for both the encoder and decoder blocks that are much deeper.



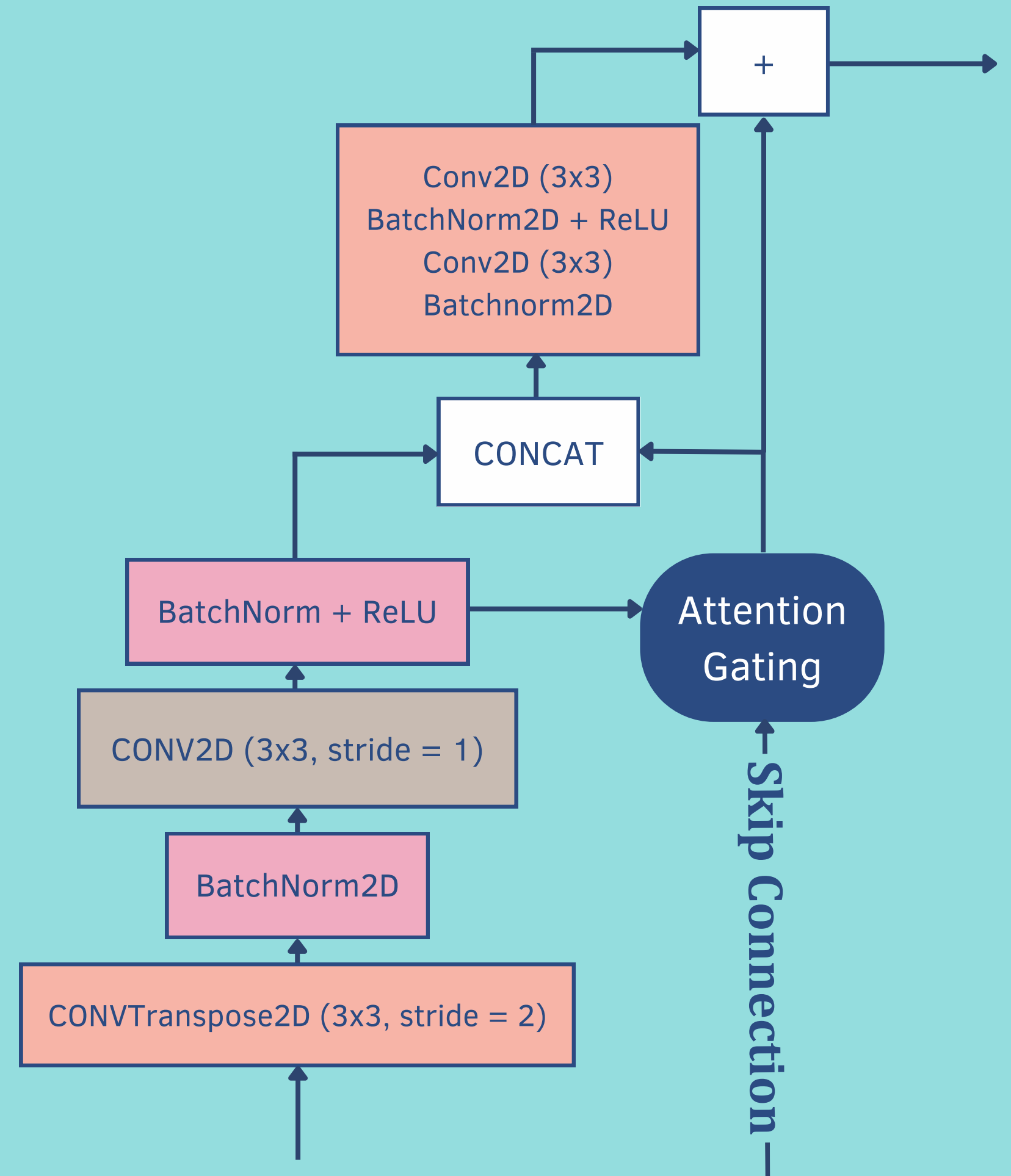
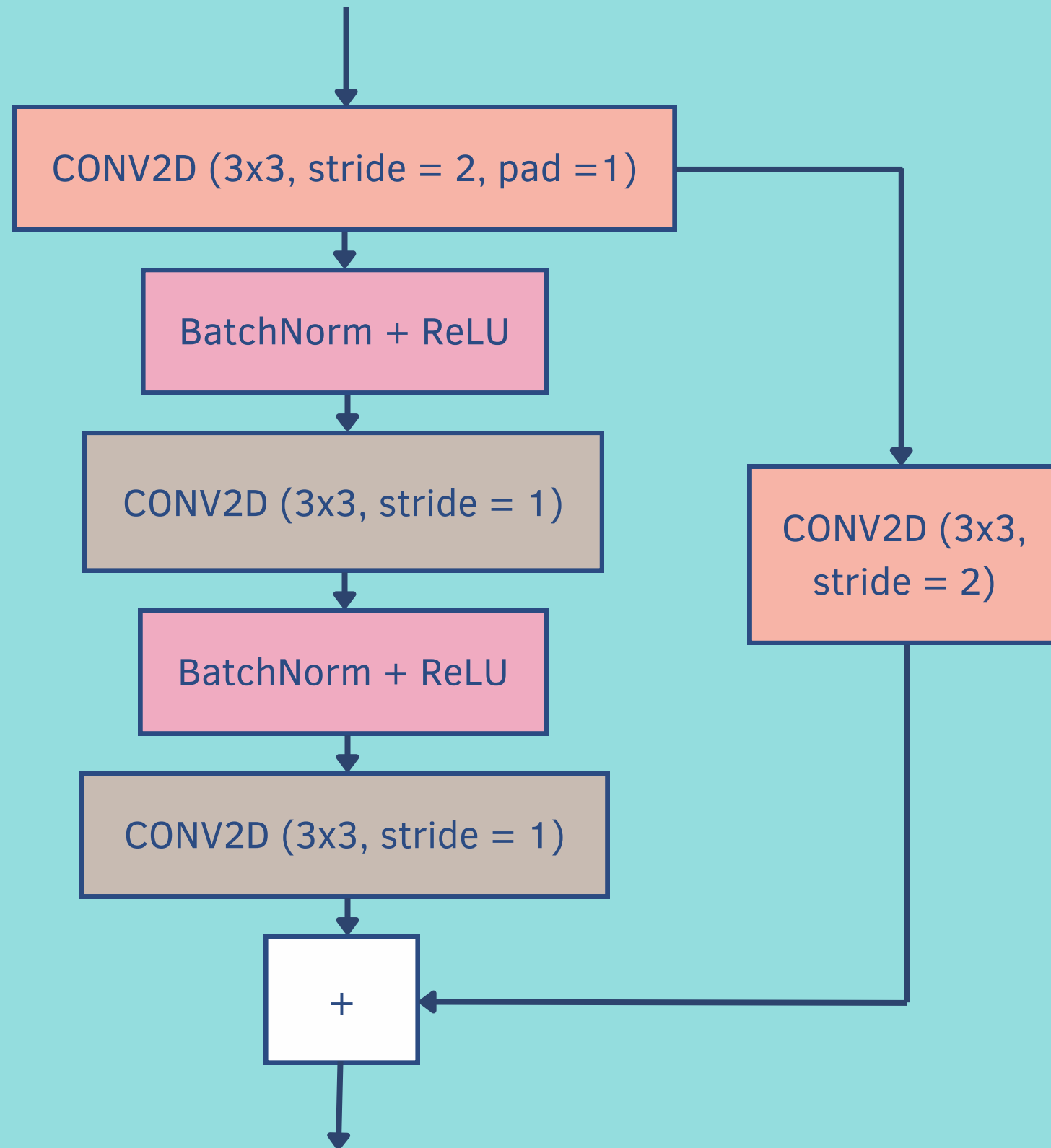
Ground Truths



Model Outputs



Custom Residual Blocks

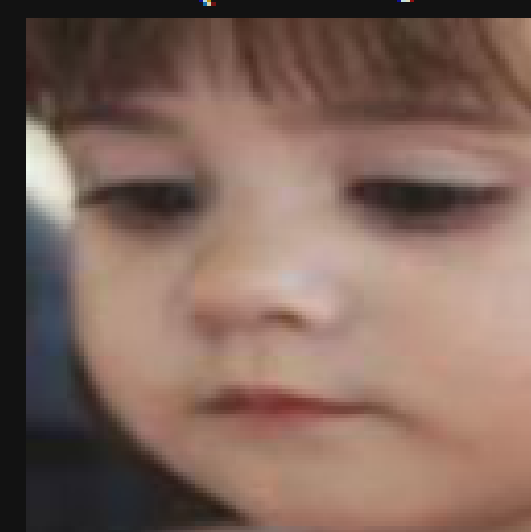
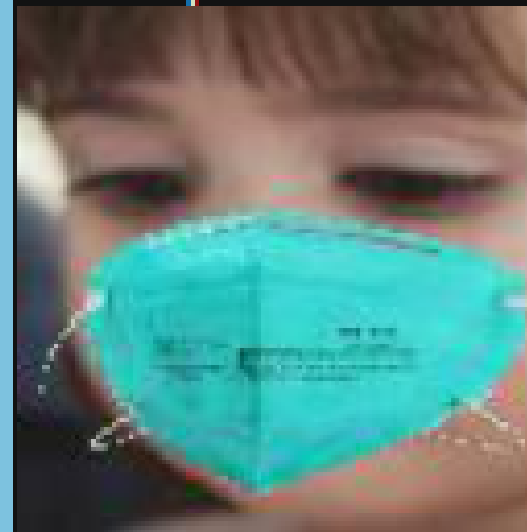


RESULTS OF RAC-UNET

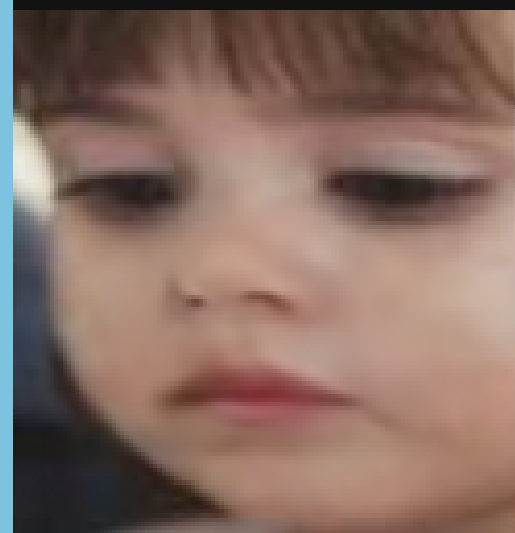
Equipped with Residual Blocks at all layers along with the conditional input and attention blocks, we can clearly see as expected that the RAC-UNET is much better for this task.

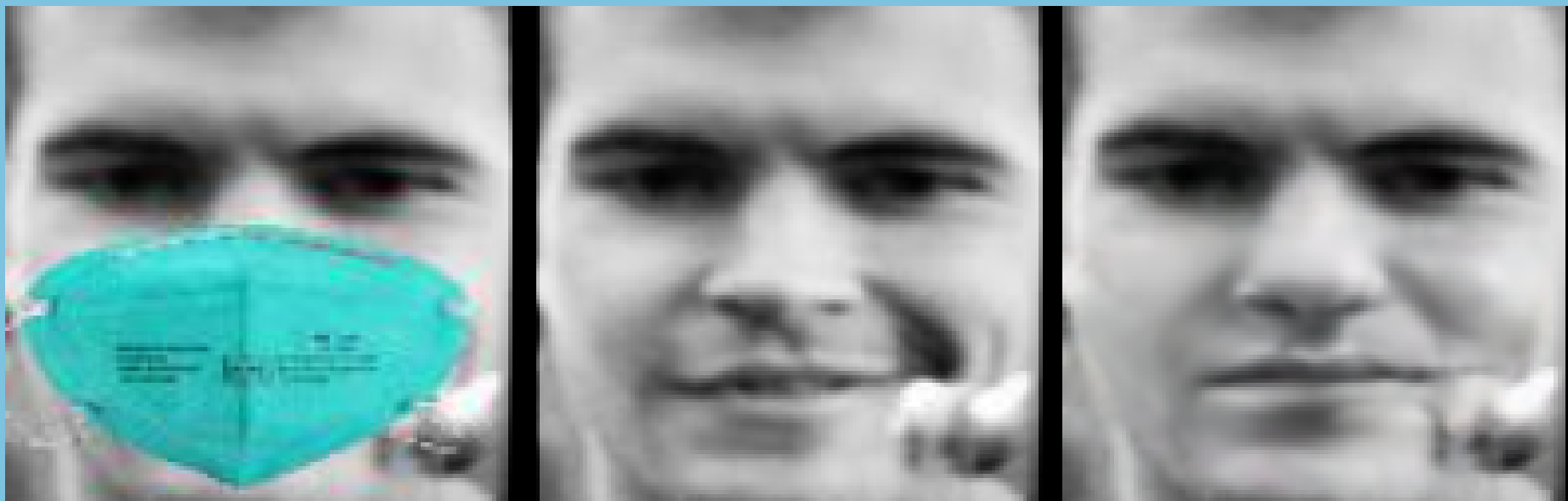
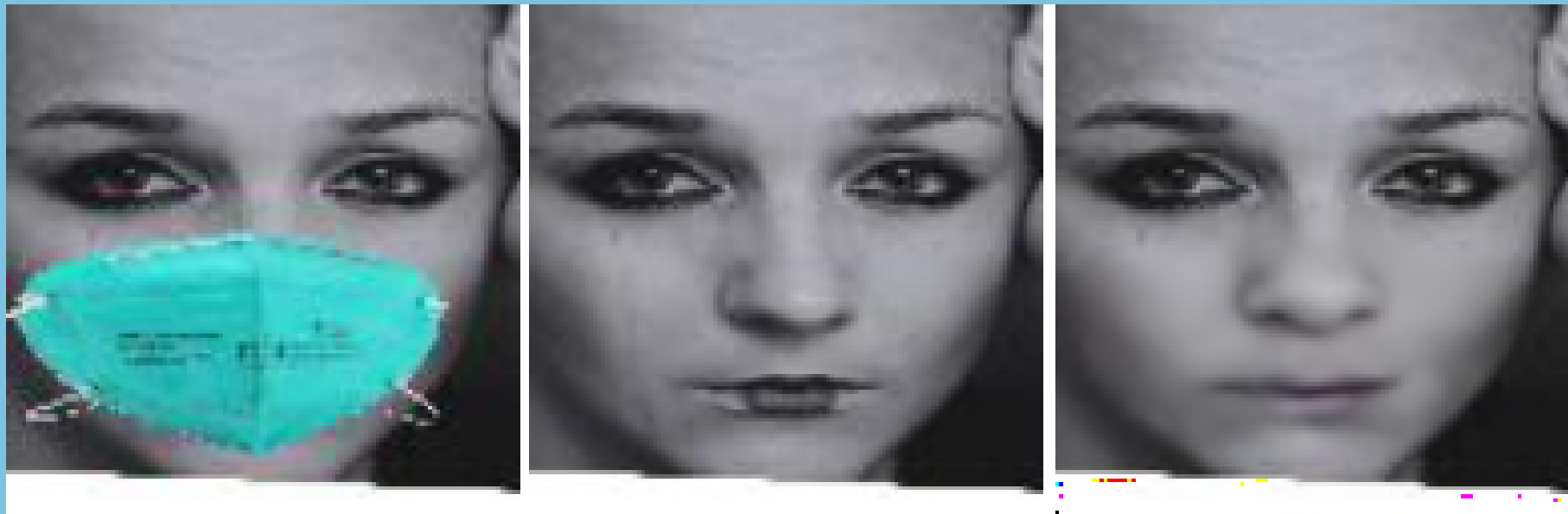
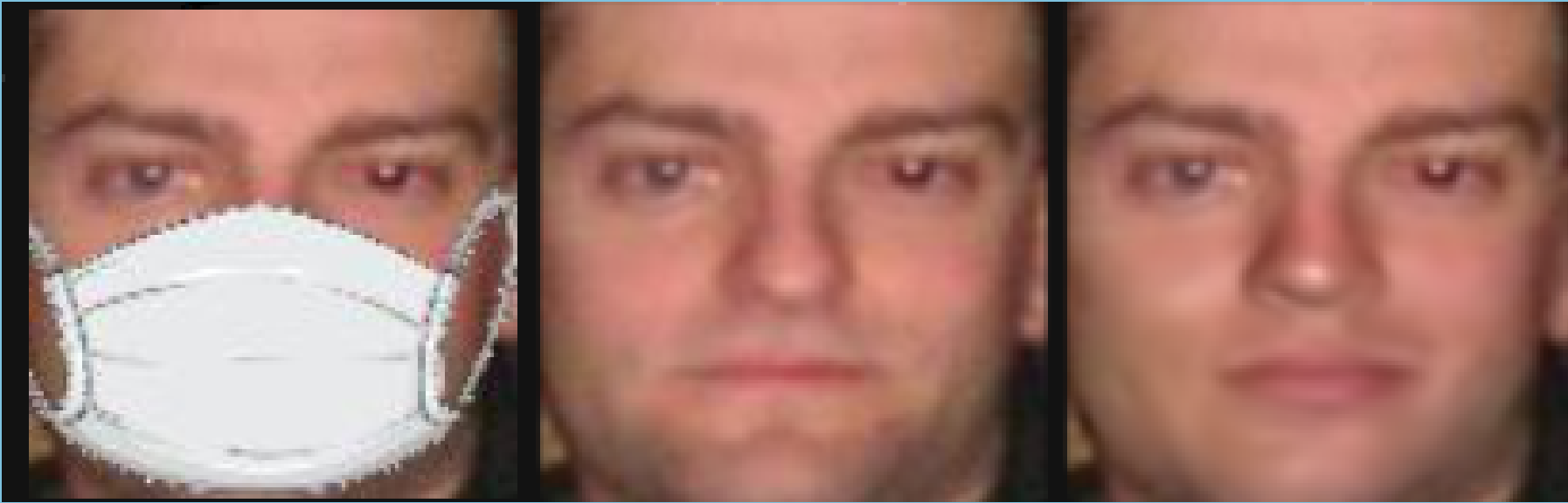
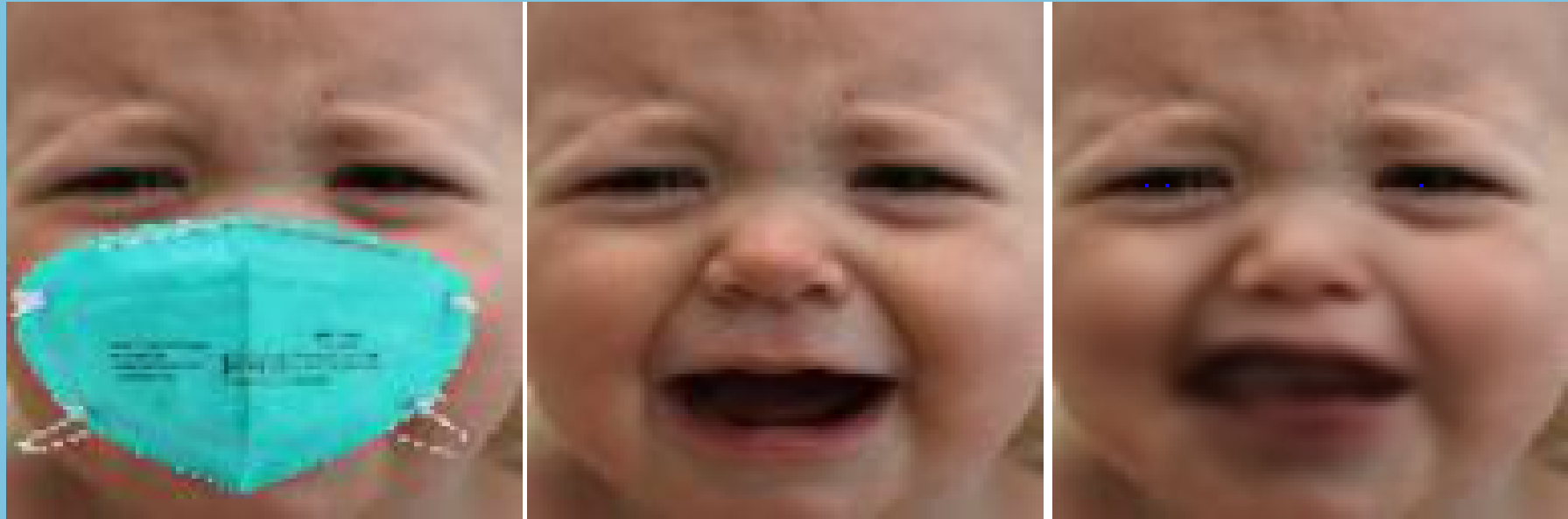
The choice of the UNET was mainly to develop a simple model that was easy to train.

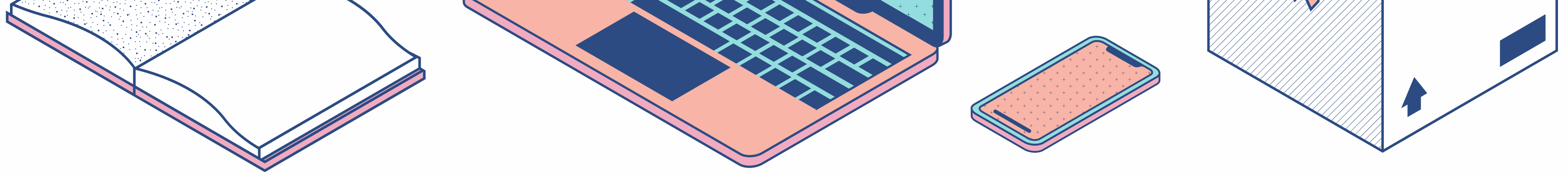
```
Model: RAC-UNET RAFDB epoch250 with MS-SSIM  
Time to compute: 0.011640259999694536 on cuda  
Emotion label: tensor([6], device='cuda:0')  
PSNR: tensor(30.4822, device='cuda:0')  
MS_SSIM: tensor(4.0150, device='cuda:0')  
Perceptual Loss: tensor(0.5466, device='cuda:0')
```



Prediction



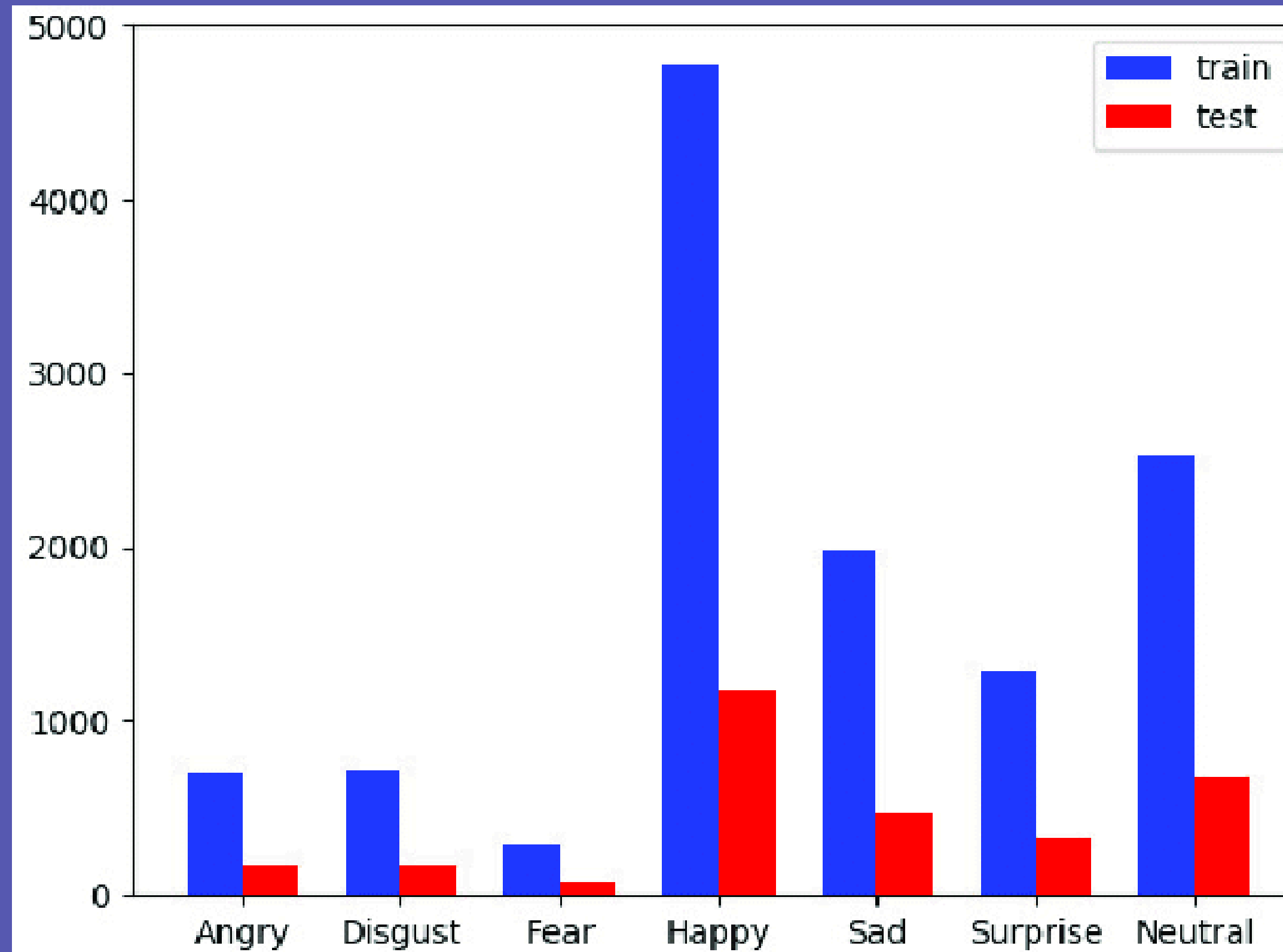




Limitations

- Size of RAFDB and depth of network resulted in overfitting.
- The Emotion distribution of RAFDB is very unbalanced.
- Model might be unnecessarily large.
- The dataset was then augmented with samples from AffectNet to get 35000 samples for training.
- The emotion distribution was balanced out to 5000 samples each.
- Methods like Quantization and Pruning can be applied as a part of future work.

Data distribution of RAFDB classes (Shows a long-tailed distribution)



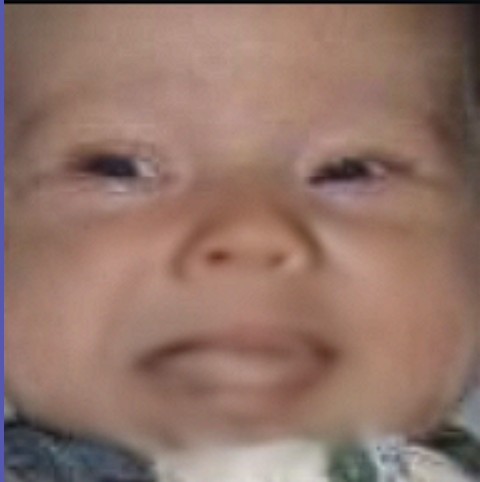
Bad prediction example

Masked

```
Time to compute: 0.007364278000068225 on cuda  
Loss: tensor(39.1410, device='cuda:0')  
Emotion label: tensor([3], device='cuda:0')  
PSNR: tensor(26.8198, device='cuda:0')
```



Prediction



Validation and Testing

A number of ablation studies were conducted to verify the robustness of the model.

- Metric value comparisons:
 - MS-SSIM, VGGPerceptual Loss, PSNR, Prediction time, BRISQUE.
- Visual inspection for blurriness, disfigurations etc.
- Expression Classification of predictions using a RESNET-18 classifier (Confusion matrix).



A Brief note on the Metrics

MS-SSIM(Multi Scale Structural Similarity Index)

MS-SSIM is a perceptual similarity metric that measures the structural similarity between two images at multiple scales. It takes into account luminance, contrast, and structural information. During training MS-SSIM encourages the generated images to have similar structural characteristics to the target images.

PSNR(Peak Signal-To-Noise Ratio)

PSNR is a commonly used objective metric for measuring the quality of reconstructed or compressed images or videos.

BRISQUE

(Blind/Referenceless Image Spatial Quality Evaluator)

Extracts a set of statistical features from an input image and comparing them to a pre-trained model. These features capture statistics related to natural images, such as contrast, sharpness, texture, and noise. Doesn't need a GT to compare with.

VGG Perceptual Loss

VGGPerceptual loss utilizes a pre-trained VGG network to extract features from the generated and target images. It measures the perceptual difference (Euclidean distance) between images. By considering higher-level features, such as textures, colors, and object shapes, it encourages the generated images to match the target images in terms of visual appearance.

Comparison of the Models

METRIC WISE COMPARISON OF THE DIFFERENT MODELS
VALUES SHOWN ARE IMAGE-WISE FOR TEST-SET OF RAFDB



	RA-UNET	RAC-UNET WITHOUT EMOTIONAL EMBED	RAC-UNET MS-SSIM	RAC-UNET PERCPETUAL LOSS	MS-SSIM + PERCEPTUAL	PERCEPTUAL + MS-SSIM
PSNR	3.0691	3.0696	3.0508	3.0297	3.0545	2.9867
MS-SSIM	0.8883	0.8732	0.8809	0.8948	0.8974	0.9186
Perceptual Loss	0.1098	0.1090	0.1115	0.1015	0.1087	0.1053
BRISQUE	50.244	48.47	44.329	46.703	45.7393	38.5344
Correct Predictions	133	1116	138	1120	1104	1147

Comparison of the Models

PERFORMANCE COMPARISON OF THE DIFFERENT MODELS TRAINED ONLY ON RAFDB

MASKED IMAGES

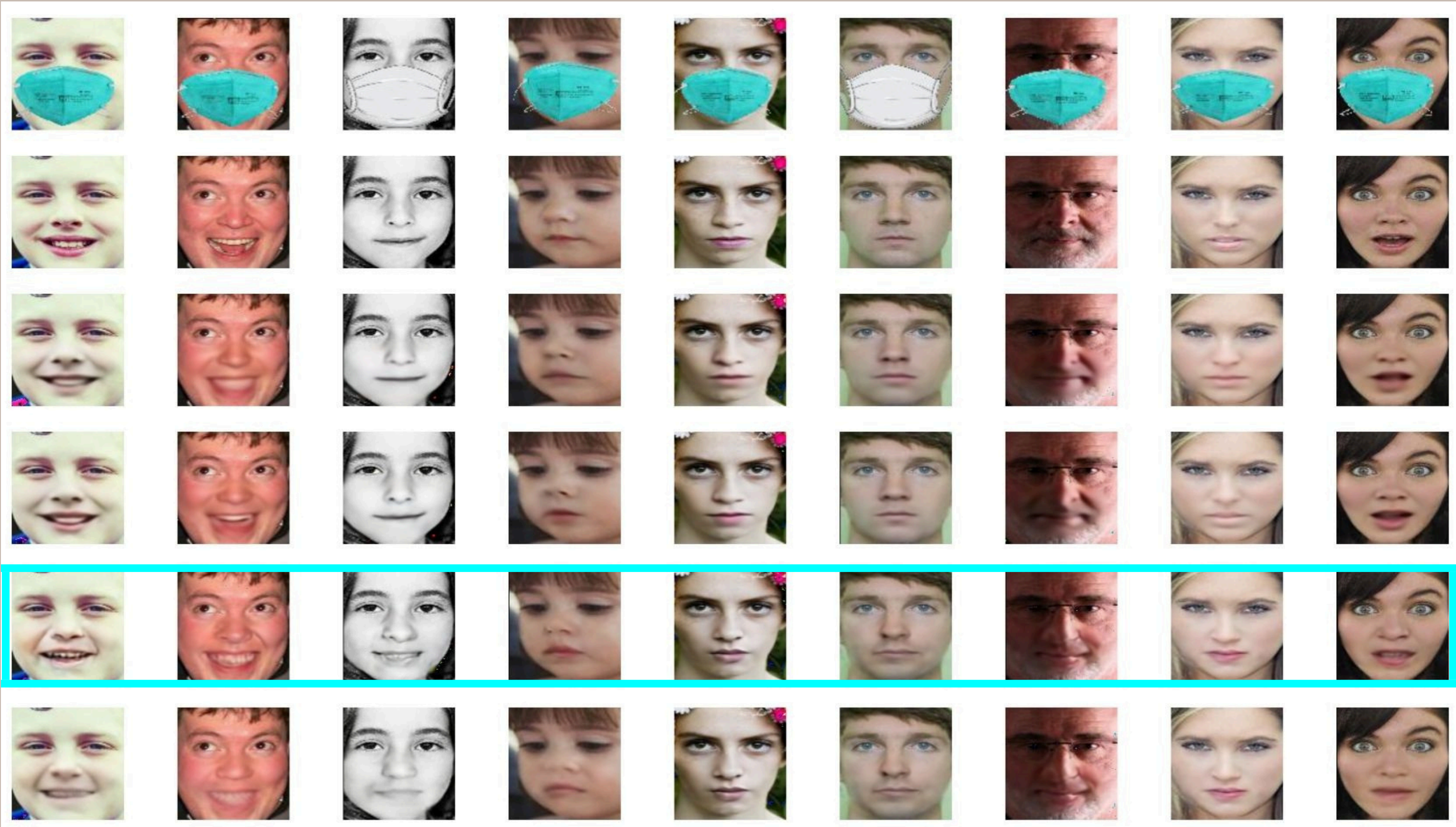
GROUND TRUTH

A-UNET

RA-UNET

RAC-UNET
(MS-SSIM)

RAC-UNET
(PERCEPTUAL)





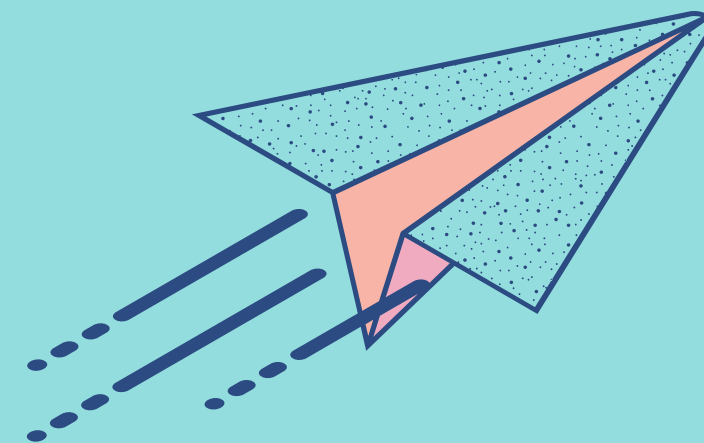
More Ablation Studies

Emotional classification of the Predictions

A Resnet - 18 based Facial Expression Classifier was trained on RAFDB + AffectNet (80.5% accuracy on only RAFDB) to classify the emotions of predictions.

RAFDB + AffectNet

The RAC-UNET model was trained from scratch on a combination of RAFDB and AffectNet to balance the emotional distribution (~5000 samples each).

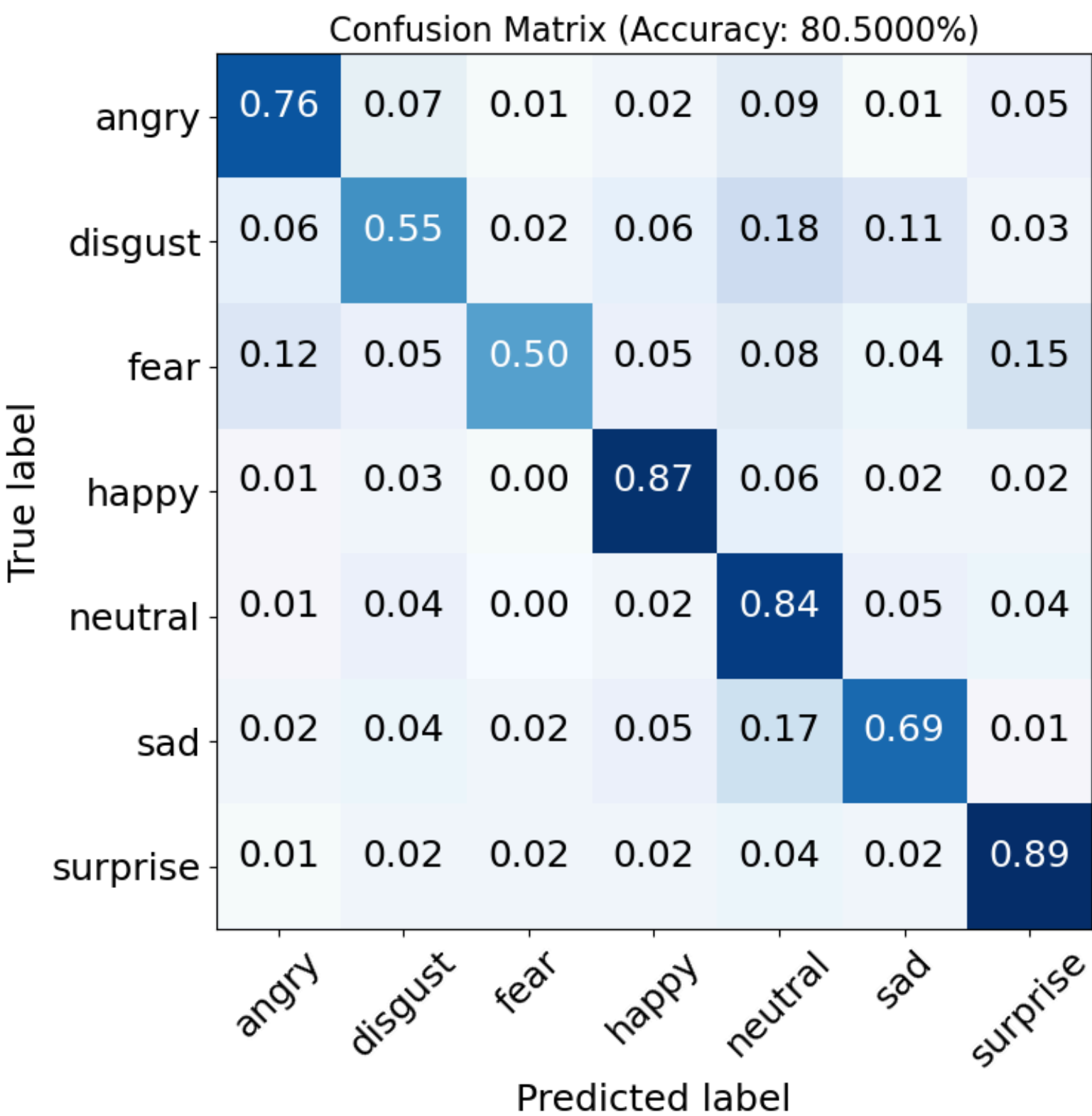


RESNET-18 Classifier

A Resnet-18 based Facial Emotional Classifier by Dr. Darshan Gera et al (DNFER) was trained on RAFDB + AffectNet.

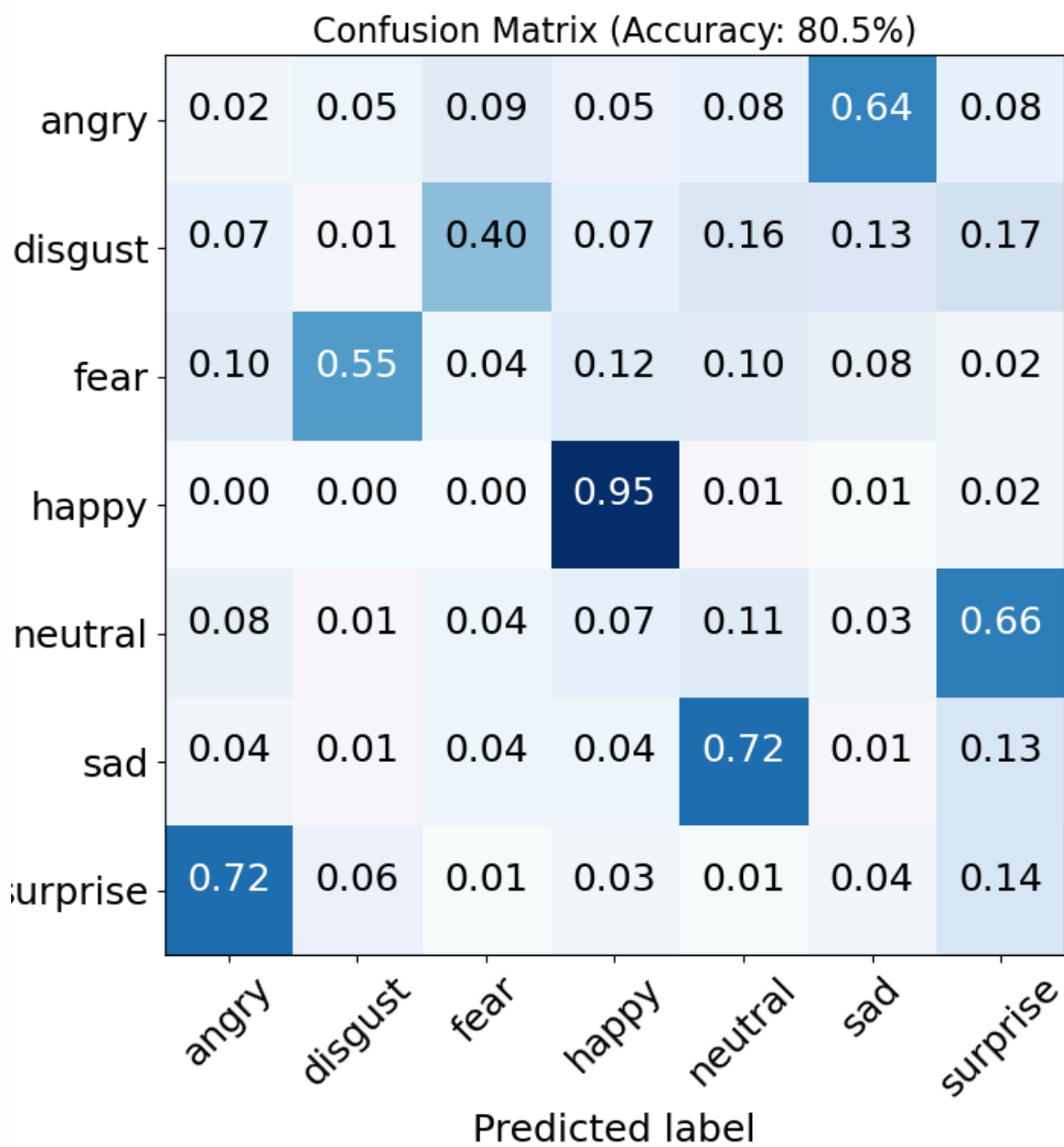
Final FER accuracy - 80.5% (On RAFDB test).
(Actual accuracy by author ~ 95%)

We take the outputs of the models and classify them with DNFER and plot the confusion matrix, using the true label of the ground truth.

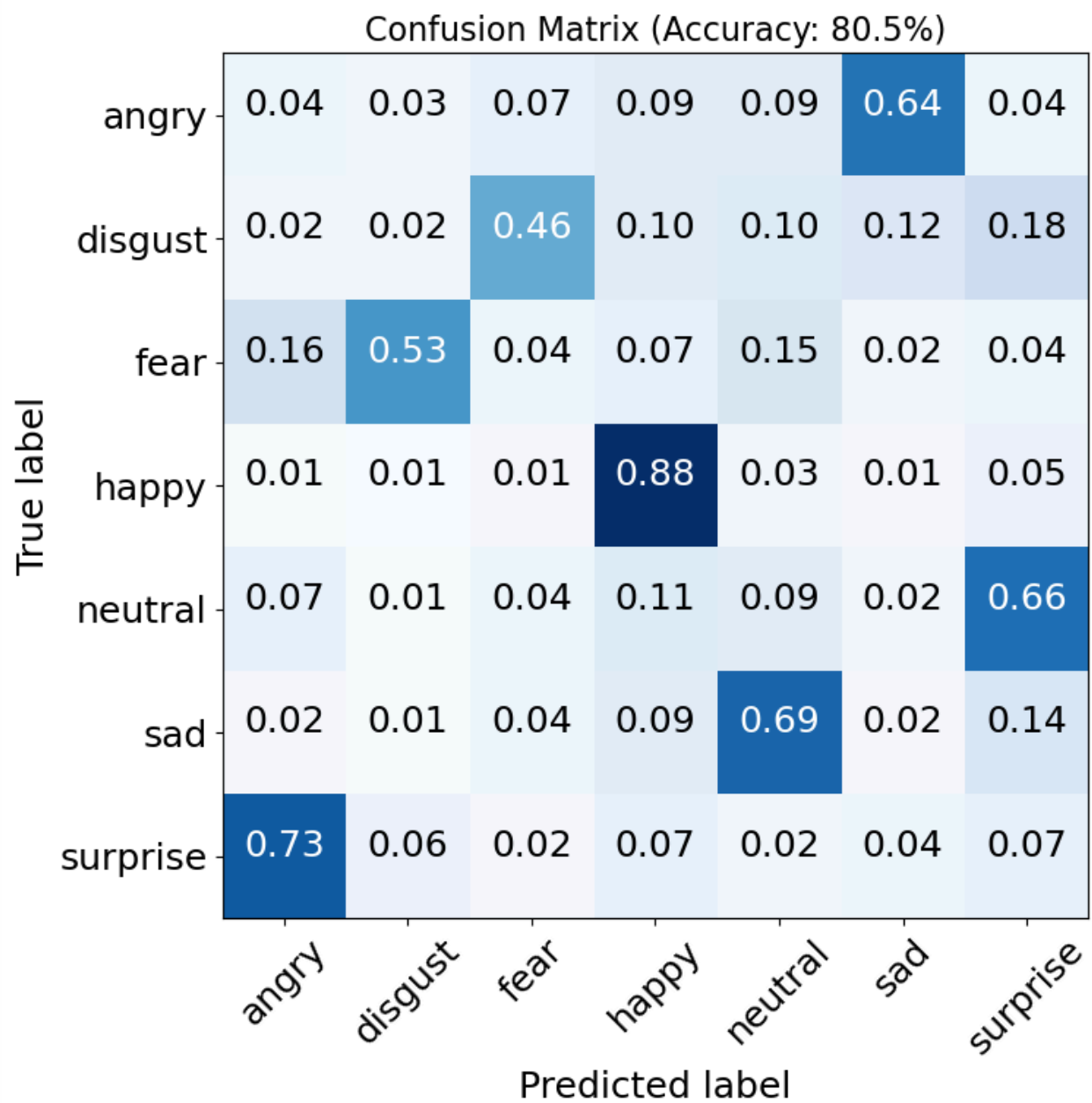


Confusion Matrix of DNFER on RAFDB test

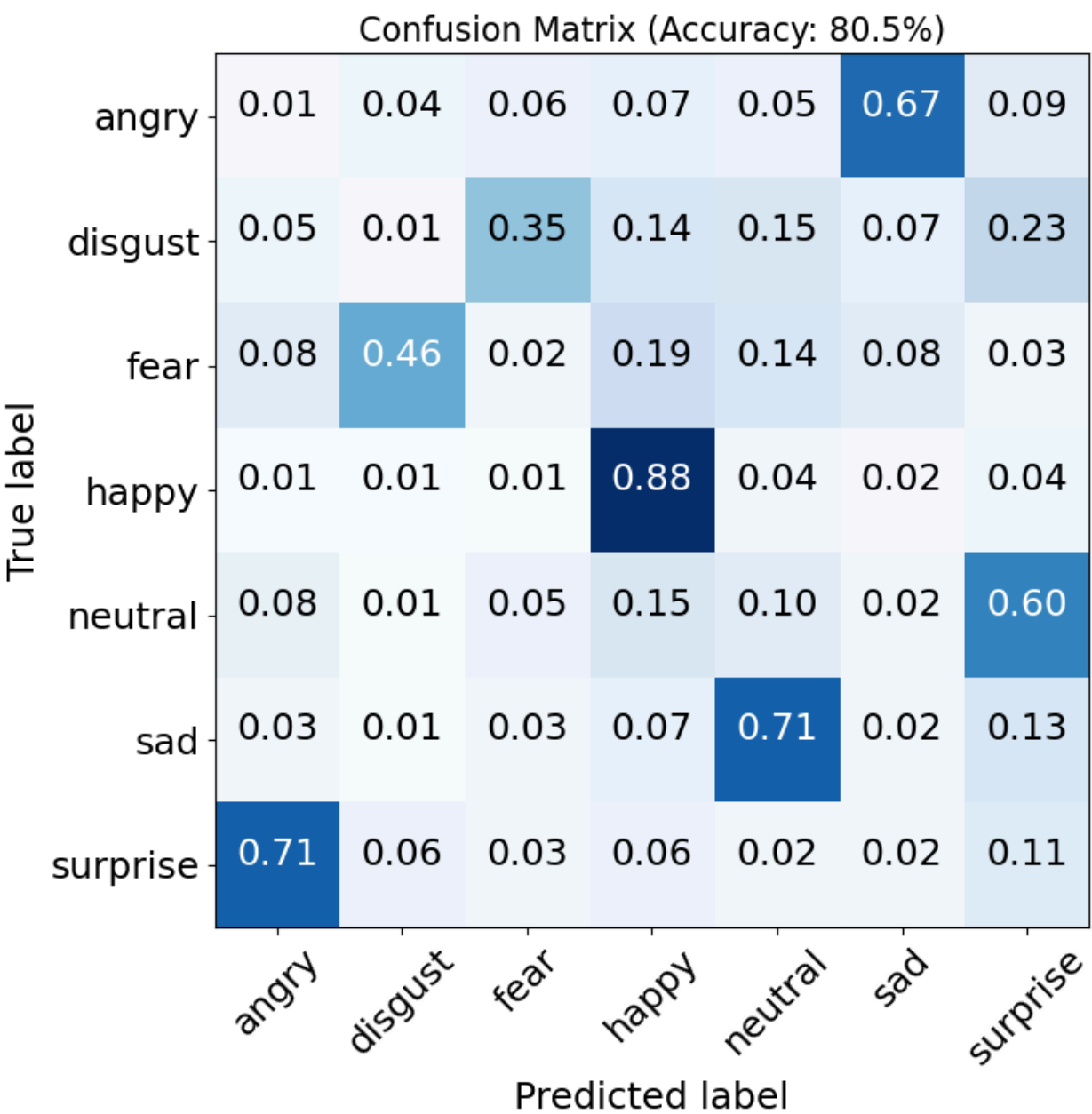
Confusion Plots of the Different Models



RAC-UNET trained on
RAFDB + AffectNet



RAC-UNET trained on
RAFDB alone



RA-UNET (Md. Hosen)
trained on RAFDB alone

FER Classification of the Outputs

Yes, the model failed...

In terms of replicating the emotion of the subject, the model has definitely failed. This can be majorly because of the imbalance in the emotional distribution.

This is why the next study is important...

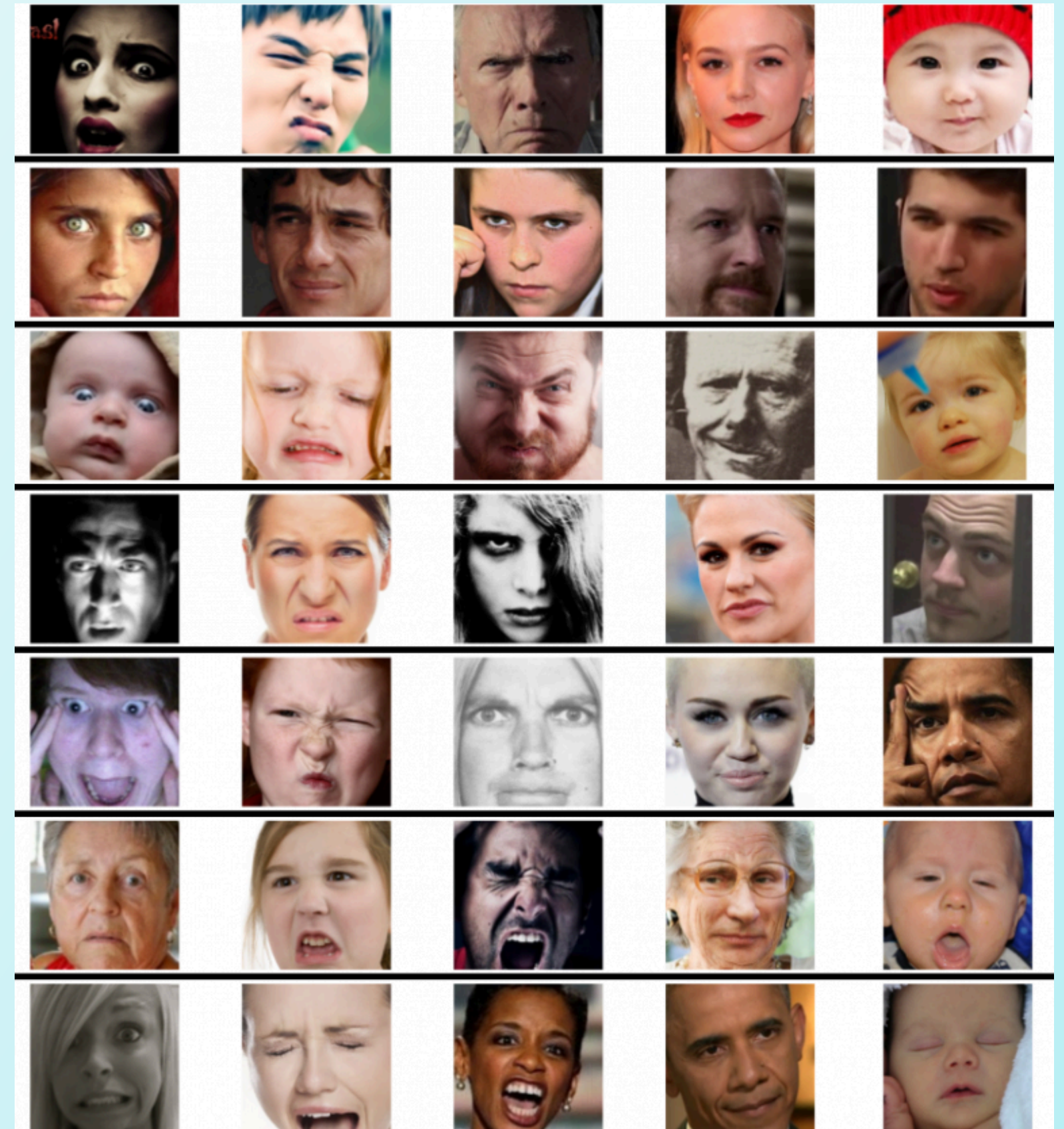
Also, there is no loss function that tells the model the difference between the emotions.

	NO. OF SUCCESSFUL EMOTION REPLICATION
RA-UNET	133
RAC-UNET WITHOUT EMOTIONAL EMBED	1116
RAC-UNET MS-SSIM	138
RAC-UNET PERCPETUAL LOSS	1120
MS-SSIM + PERCEPTUAL	1104
PERCEPTUAL + MS-SSIM	1147

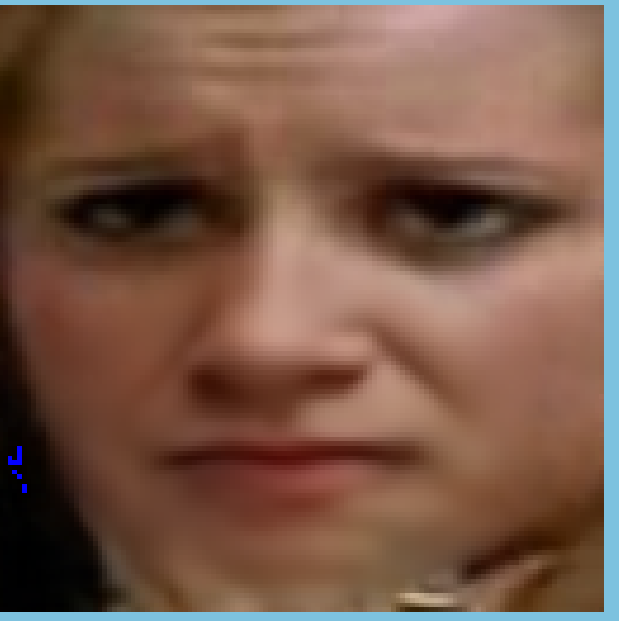
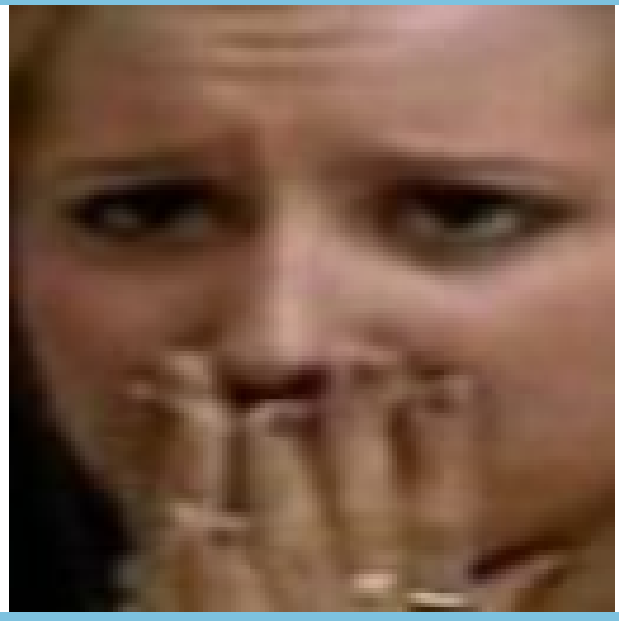
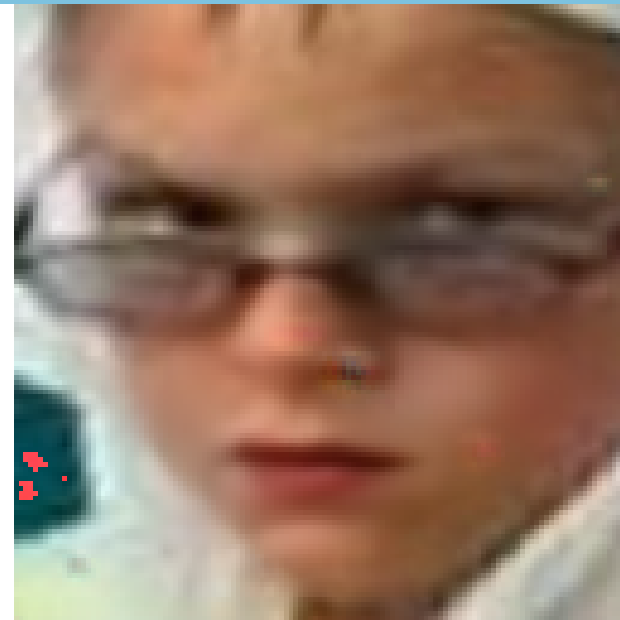
RAFDB + AffectNet

The performance of the model increased significantly when the RAFDB trainset was augmented with samples of AffectNet, to give a balanced emotional distribution.

The model still blindly learns the differences between emotions but it is enough to increase the no. of correct outputs by a staggering amount.



	RA-UNET	RAC-UNET MS-SSIM	RAC-UNET PERCEPTUAL	RAC-UNET RAFDB + AFFECTNET
PSNR	3.0691	3.0508	3.0297	3.0294
MS-SSIM	0.8883	0.8809	0.8948	0.9569
Perceptual Loss	0.1098	0.1115	0.1015	0.114
BRISQUE	50.244	44.329	46.703	43.0968
No. of correct Predictions	133	138	1120	1208

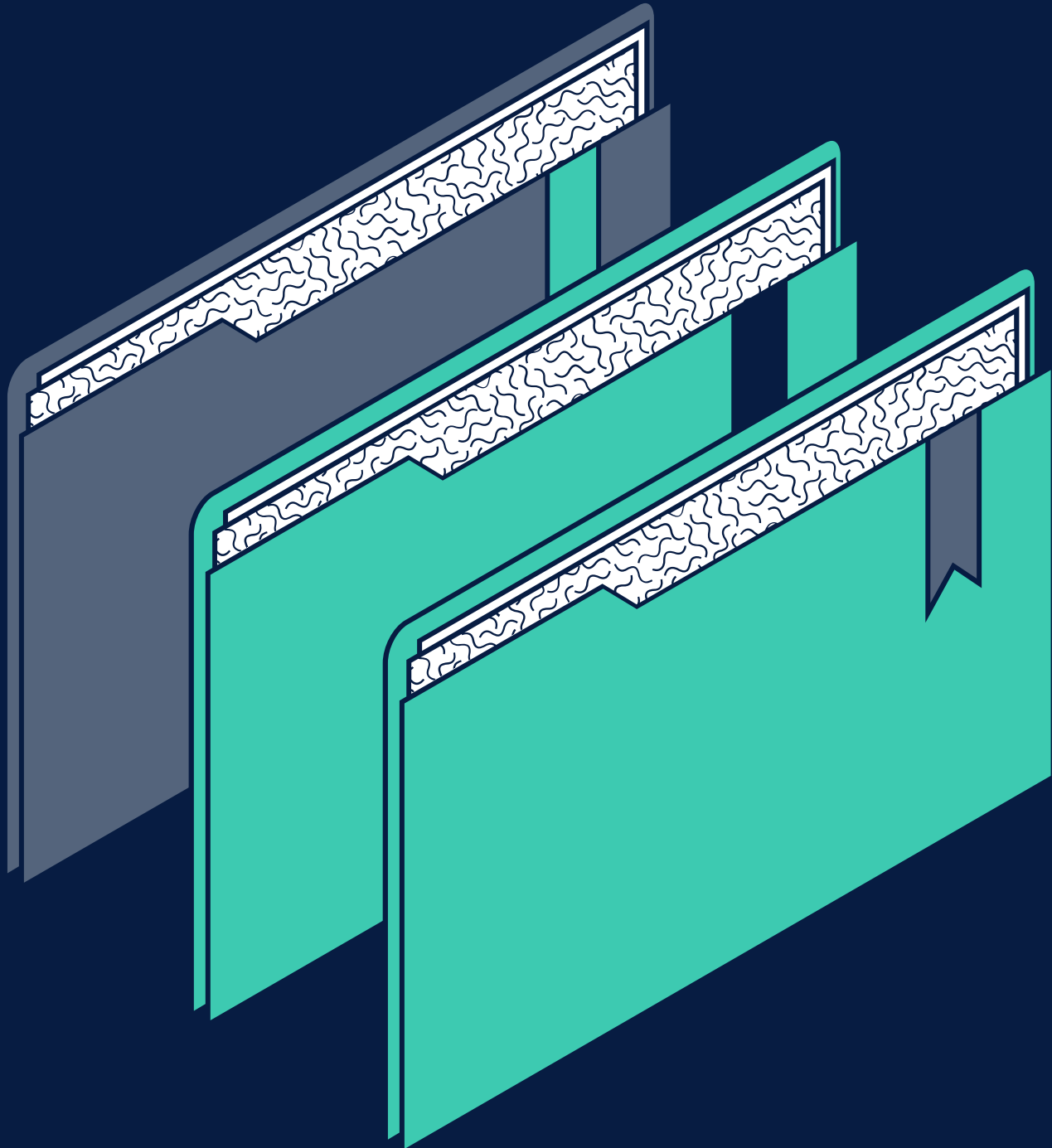


The Road Forward...

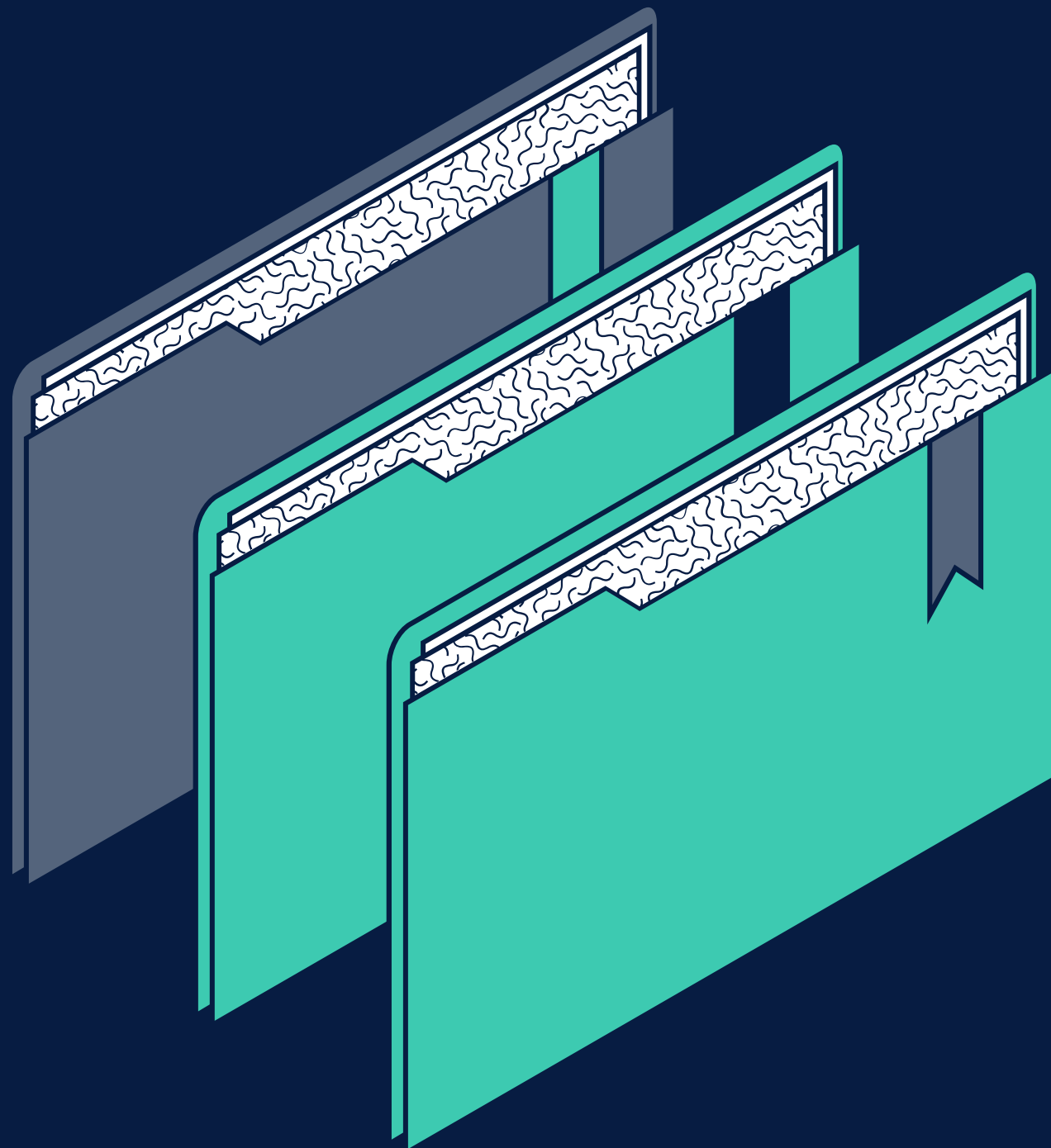


- Implement a loss function the model can use and learn to in-paint different emotions.
- Use Binary segmentation to detect the masked region and in-paint accurately.
- Reduce the model size and complexity using Knowledge Distillation, Quantization etc.
- Fit the model to in-paint on images with actual face masks.

References



- **Inpainting using Residual Attention U-Net**
<https://arxiv.org/pdf/2209.08850.pdf>
- **STRV Blog**
<https://www.strv.com/blog/mask2face-how-we-built-ai-that-shows-face-beneath-mask-engineering>
- **U-Net: Convolutional networks for segmentation**
<https://arxiv.org/pdf/1505.04597.pdf>
- **Unmasking Your Expression: Expression-Conditioned GAN for Masked Face Inpainting** by Sridhar Sola, Dr. Darshan Gera



References

- **Dynamic Adaptive Threshold based Learning for Noisy Annotations Robust Facial Expression Recognition:** Darshan Gera, S Balasubramanian et al.
- **Deep residual learning for image recognition**
Proceedings of the IEEE conference on computer vision, 2016. He, Kaiming, et al.
- **Attention U-Net: Learning Where to Look for the Pancreas** Oktay, Ozan, et al. arXiv:1804.03999 (2018).

Thank
you!

